

NBA Cluster Analysis

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Project Overview

- Cluster NBA players and teams by quality
- Variables: shooting percentages, assists, blocks, rebounds, etc.
- Method: PCA + Model-Based Clustering (R)
- Analysis covers mainly 2023-2024 NBA season

NBA Data

- Source: NocturneBear GitHub (2010–2024)
- Initial data: 34 variables, 424,478 observations
- Cleaned and aggregated data
- Separate datasets for players and teams

[illegible]

Figure: NBA Data

Player Variables

- Rates: FG%, 3P%, FT%
- Aggregate: minutes, assists, blocks, points, rebounds, steals, turnovers, fouls

Team Variables

- Rates: FG%, 3P%, FT%
- Aggregate: assists, blocks, points, rebounds, steals, turnovers

Player Data – Correlation Analysis

- Strong correlations due to minutes played
- Percentages independent from minutes

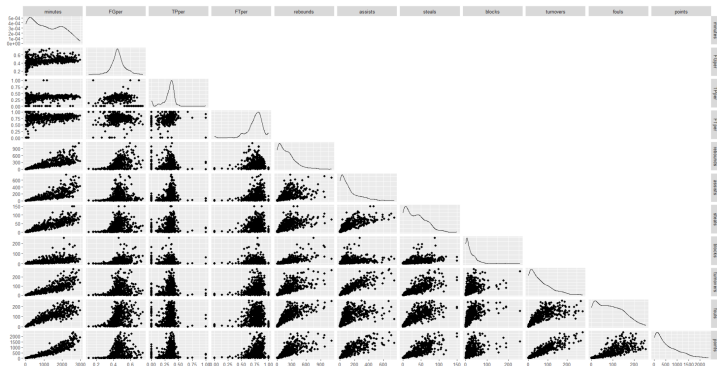


Figure: Players Pairs Plot

Regular Players

- Players with ≥ 1000 minutes (263 players)
- Reduced noise from low-minute players

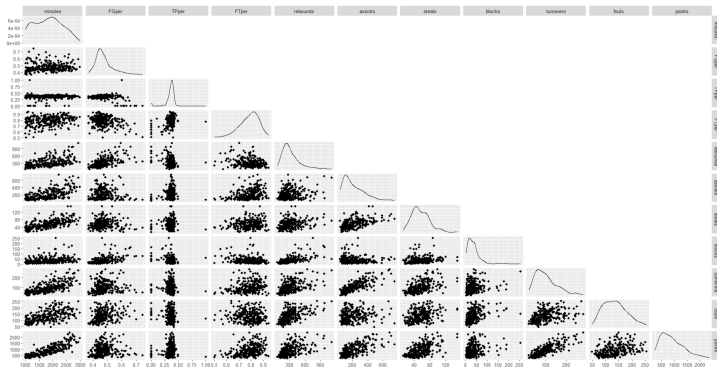


Figure: Pairs Plot After Reduction of Noise

Initial Player Clustering

- Initial run: 4 clusters suggested
- Issue: Mixed positions in clusters
- Example: Guard (Conley) grouped with Forward (Leonard)



conley2024photo



leonard2025photo

PCA and Clustering

- PCA dimension reduction ($11 \rightarrow 5$ variables)
- PCA improved clarity, positions still mixed
- Led to position-based clustering

Players Split by Position

- Separate clustering: Centers, Forwards, Guards
- Clearer positional clusters

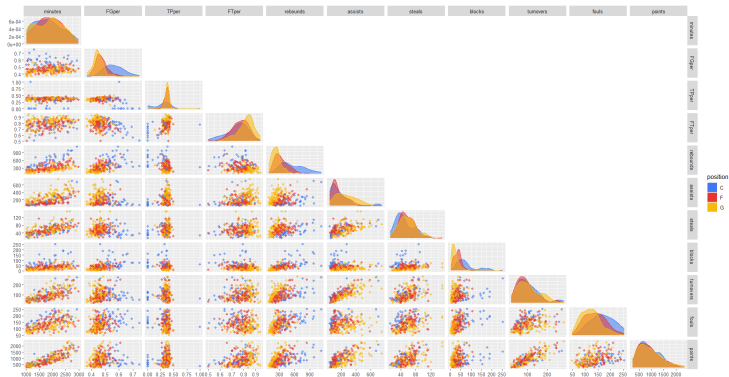


Figure: Postion Pairs Plot

Clustering Centers

- 3 clusters (Varsity, JV, B-Squad)
- Jokic, Porzingis grouped clearly
- PCA explained 83% variance



sportingnews2024jokic

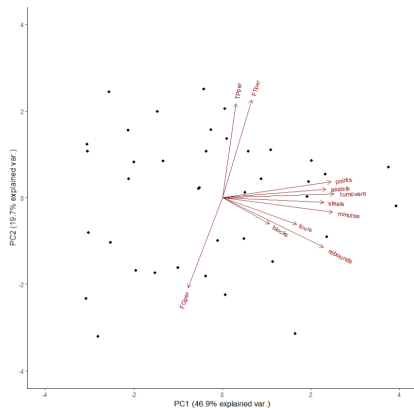


Figure: Center BiPlot

Clustering Centers cont.



Figure: Center Clusters

Clustering Forwards

- 3 clusters identified
- Cluster 1: Stars (Durant, Butler, LeBron)
- PCA explained 85% variance



durant2023photo



butlerherro2020photo



lebronfunnyface2019photo



nba2024nazreid

Clustering Forwards cont.



Figure: Forward Clusters

Clustering Guards

- Clear skill differences identified
- Doncic and Edwards notably clustered
- PCA explained 85% variance



doncic2025photo



edwards2025photo

Clustering Guards cont.



Figure: Guard Clusters

Team Data Analysis

- PCA (4 PCs): Explained 80% variance
- Less skewness compared to player data



Figure: Team Correlation Plot

Team Clusters (2023–2024)

- Initially 8 clusters; reduced to 5 for clarity
- Example: Cluster 4 aligned with playoff teams (MIN, BOS, IND)

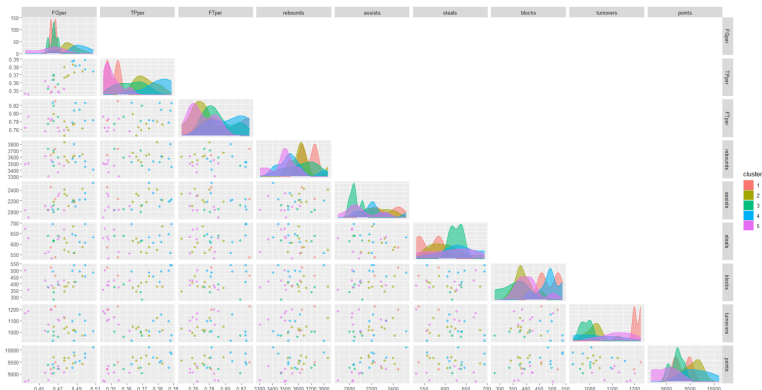


Figure: Team Clusters (2023-24)

Team Clusters (2013–2014)

- Compared 2013–2014 to recent clusters
- Minimal overlap indicates substantial team changes

Table: Team cluster classifications (2023–2024)

Cluster 1	2	3	4	5
SAS	ATL	CHI	BOS	BKN
UTA	CLE	HOU	IND	CHA
	DAL	MIA	LAC	DET
	DEN	NYK	LAL	MEM
	GSW		MIN	ORL
	MIL		OKC	POR
	NOP		PHI	TOR
	SAC		PHX	WAS

Table: Team cluster classifications (2013–2014)

Cluster 1	2	3	4	5
ATL	BOS	CHA	DEN	DET
BKN	CHI	MEM	GSW	PHI
DAL	CLE	NOP	HOU	
LAC	IND	NYK	LAL	
MIA	MIL		OKC	
MIN	ORL		PHX	
WAS	SAC		POR	
	TOR		SAS	
	UTA			

Model Applications

- Useful for trade considerations, drafting, training
- Teams could identify improvement strategies
- Adapting models to college prospects

Improvements and Future Directions

- Revisit percentage vs. aggregate statistics
- Use averages rather than season totals
- Include player salary for cost-effectiveness

Discussion and Summary

- Models aligned with expectations
- Surprising cluster size distributions
- Importance of positional differences

Thank you!